

# Make Your Own Tor Proxy Router with Raspberry Pi

Many of us want to browse anonymously on our laptops, mobile phones and tablets, or on console devices that not only do not have an Ethernet/LAN connection but also cannot run Tor. What if you cannot install Tor or do not want to on your work computer? If you want to use Tor but don't have the time or ability to run it on your computer, this article is for you.

**T**or, more commonly known as The Onion Router, allows you to anonymously use the Internet by concealing your real identity. It protects you from any traffic analysis as well as network spying. The Tor network encrypts the data into multiple layers and hides the origin of a Web page request. As you know, every Internet packet goes through three layers of relays before reaching its destination. This makes it much harder for the server being accessed to figure out the identity and origin of the request. It is an excellent way to gain access to certain websites that are blocked.

## Prerequisites

You'll need a few things to make you own Tor proxy router:

- Raspberry Pi model B+ (or B) - Ethernet
- Ethernet cable
- Wi-Fi adapter
- SD card (4GB or greater) with Raspbian on it. You can either copy the Raspbian image onto it or buy a ready-made Raspbian card
- Power supply for your Pi
- USB Console cable (optional)—this makes it a little easier to debug the system
- A case for your Pi (optional)
- An SD or microSD card reader (optional)

## Setting up Raspberry Pi

When you buy a Raspberry Pi, it does not necessarily come with an SD card. The SD card is important because this is where the Raspberry Pi keeps its operating system, and also where you will store your documents and programs.

Even if your Raspberry Pi comes with an operating system, it is always good to update it to the latest version. Remember, putting the operating system in the Raspberry Pi will wipe all the data. Hence, it is advised to have a USB drive for your documents, so that when you install a new version of the operating system, you don't have the complication of copying them off somewhere safe before reformatting the SD card.

If your Raspberry Pi did not have an SD card, then the

minimum size you should get is 4GB; but get a bigger one if you think you will need the space. It is always advisable to get a bigger size SD card.

## Downloading the image

The operating system that will be installed onto the SD card must be downloaded from the official Raspberry Pi website. This will usually be a Zip file, which then extracts to an image file of type *.img*.

Whatever image file you download, the actual installation process will be the same.

There are numerous OSs that you can install on your Raspberry Pi. If you are a beginner, you'll probably want to stick with one of the Linux distributions, but which one should you choose? Being an open source OS, you can take one of the existing distributions and add things to it or configure it in a certain way before packaging it up as another distribution option for anyone to use.

This is how the most common Raspberry Pi distribution, 'Raspbian', came into existence. After downloading the *.zip* Raspbian file, unzip it to get the image file (*.img*) for writing to your SD card.

## Writing the image to the SD card

Please visit the official Raspberry Pi website, to get the correct procedure.

1. Linux: <https://www.raspberrypi.org/documentation/installation/installing-images/linux.md>
2. OSX: <https://www.raspberrypi.org/documentation/installation/installing-images/mac.md>
3. Windows: <https://www.raspberrypi.org/documentation/installation/installing-images/windows.md>

## Bootling and configuring Raspberry Pi

Once you have the OS installed on your SD card, it's time to set up your Raspberry Pi to boot it up for the first time. To do this, use a tool called *Raspi-Config*, which automatically runs when you boot your Raspberry Pi for the first time. This starts before the operating system, and so you have to use the cursor keys and the *Enter* key to navigate through the menu system.

